

Digital Egypt Pioneers Initiative (DEPI)

DevOps Project
by
Group - 1

Ahmed Nabil Zin EL-Din

Mohamed EL Mahdy

Ahmed Hisham

Mohamed Alaa

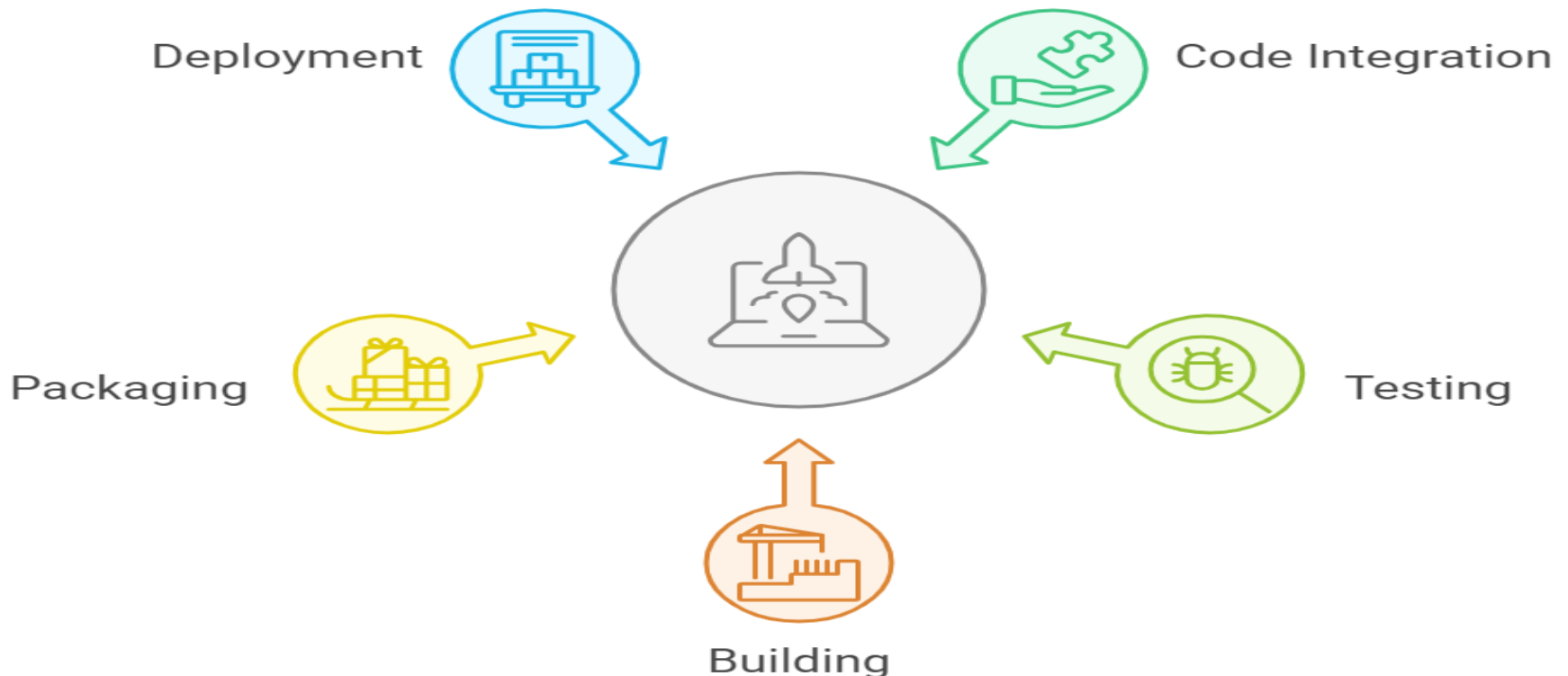
CI/CD Pipeline with GitHub Actions and Modern DevOps Tools

Overview of stages and tools used in the CI/CD
pipeline

What is CI/CD?

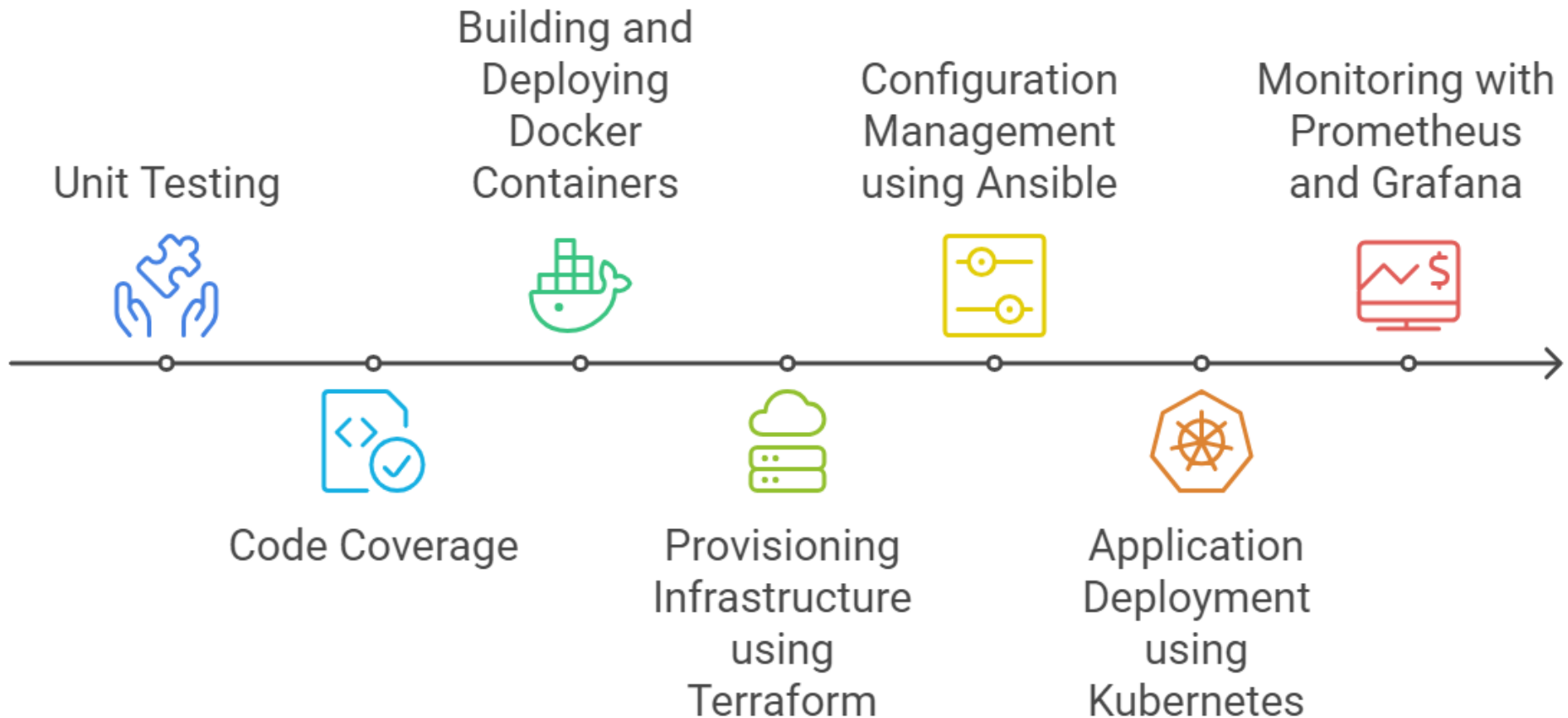
CI/CD stands for Continuous Integration and Continuous Deployment. It automates the process of integrating code changes, running tests, building, and deploying applications. This ensures rapid and reliable software releases.

CI/CD Process for Software Releases



Overview of the Pipeline

CI/CD Pipeline Sequence



Unit Testing Stage

Unit tests are run on multiple versions of Node.js and operating systems to ensure compatibility across environments. The results are archived for review.

Unit Testing Process in CI/CD Pipeline

Initiate Unit Tests



Run Tests on Node.js Versions



Run Tests on Operating Systems



Archive Test Results



Code Coverage Stage

This stage ensures that the tests cover a sufficient portion of the code. Code coverage analysis helps maintain high-quality code.

Code Coverage Stage in CI/CD Pipeline

Initiate Code
Coverage
Analysis



Measure Test
Coverage



Evaluate
Coverage
Sufficiency

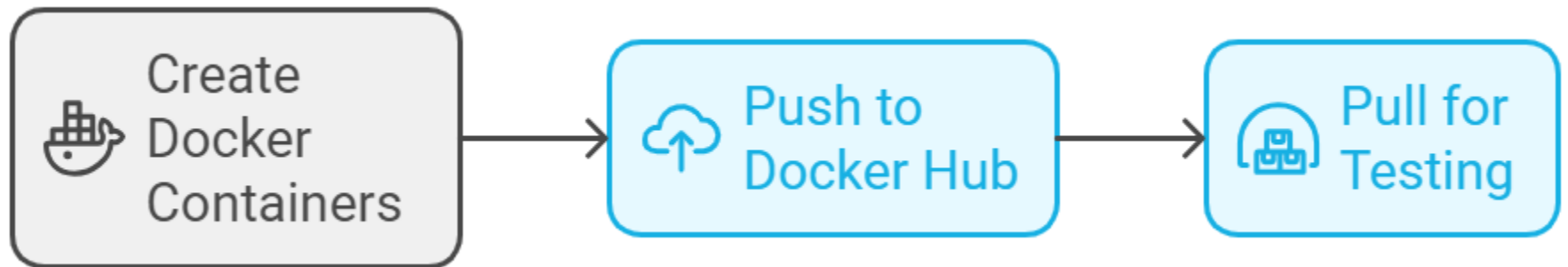


Maintain High-
Quality Code



Building and Deploying Docker Containers

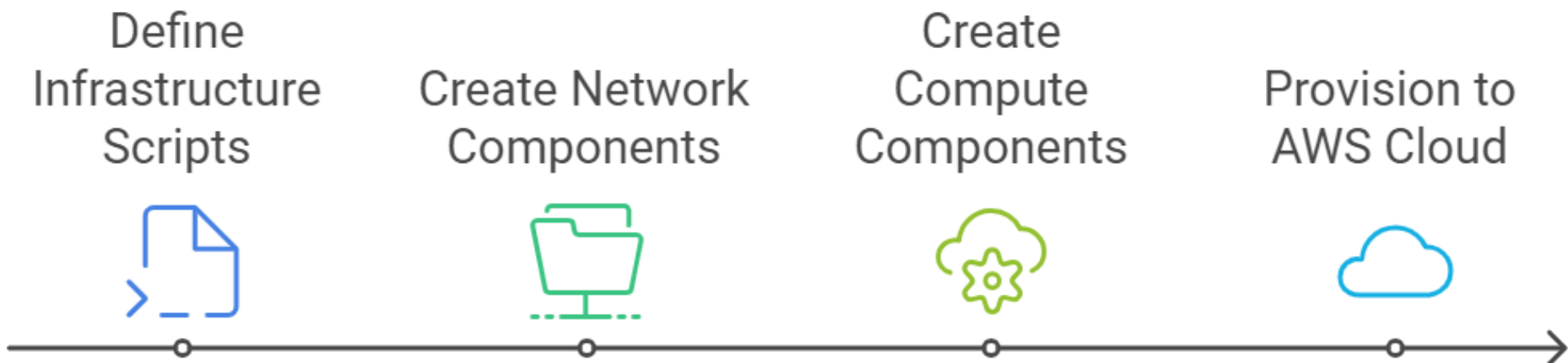
Docker is used to build containers, and to push them to Docker hub as a container registry for further testing procedures



Building Infrastructure using Terraform on AWS Cloud

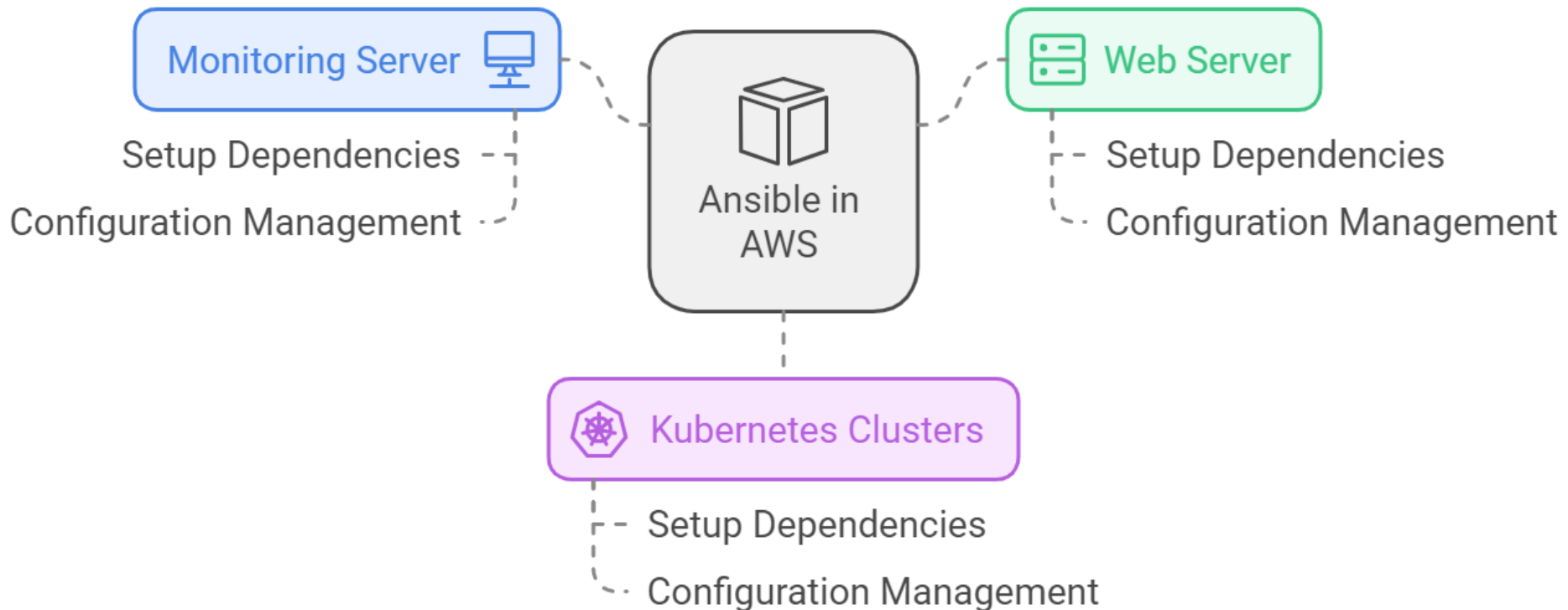
Terraform is an IaC tool to create the scripts required to build the project infrastructure including network and compute components then provisioning them to AWS Cloud platform.

Terraform Infrastructure Provisioning on AWS



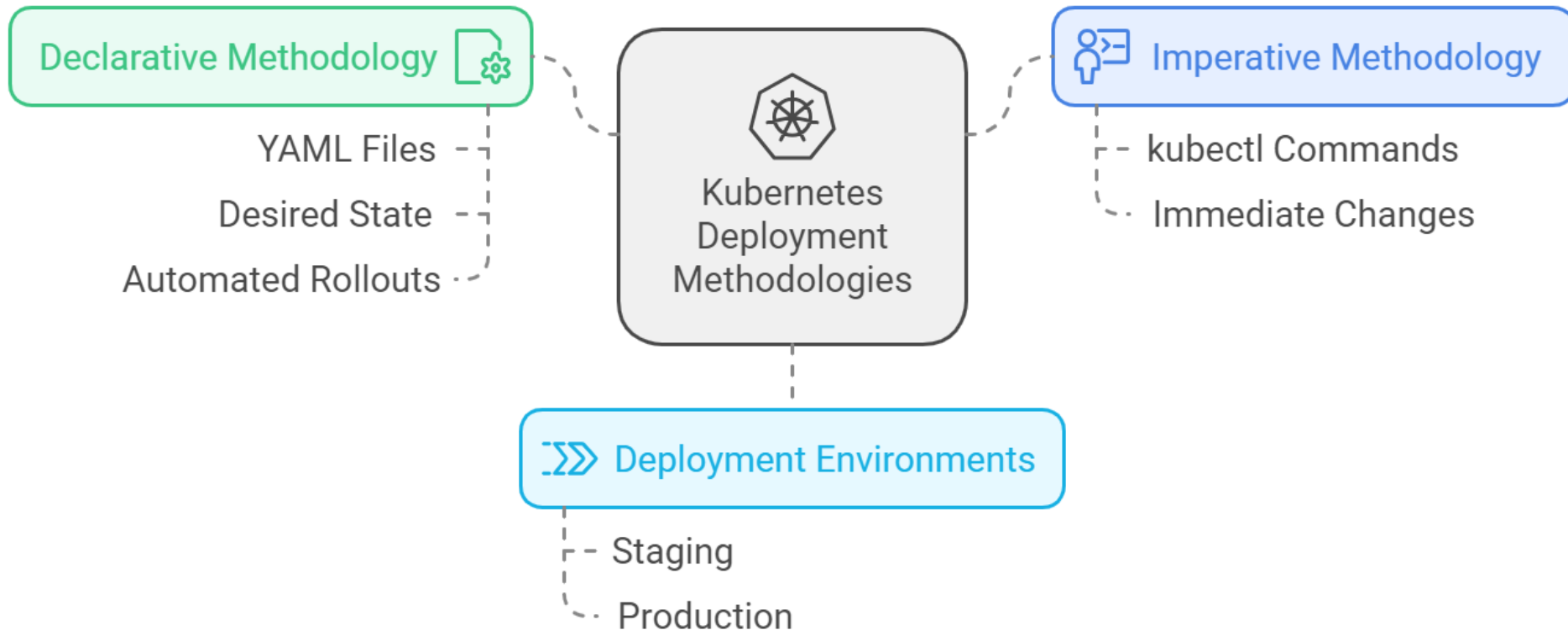
Configuration Management using Ansible

Ansible automates the setup and configuration of AWS infrastructure and Kubernetes clusters, ensuring consistent environments for the application deployment.



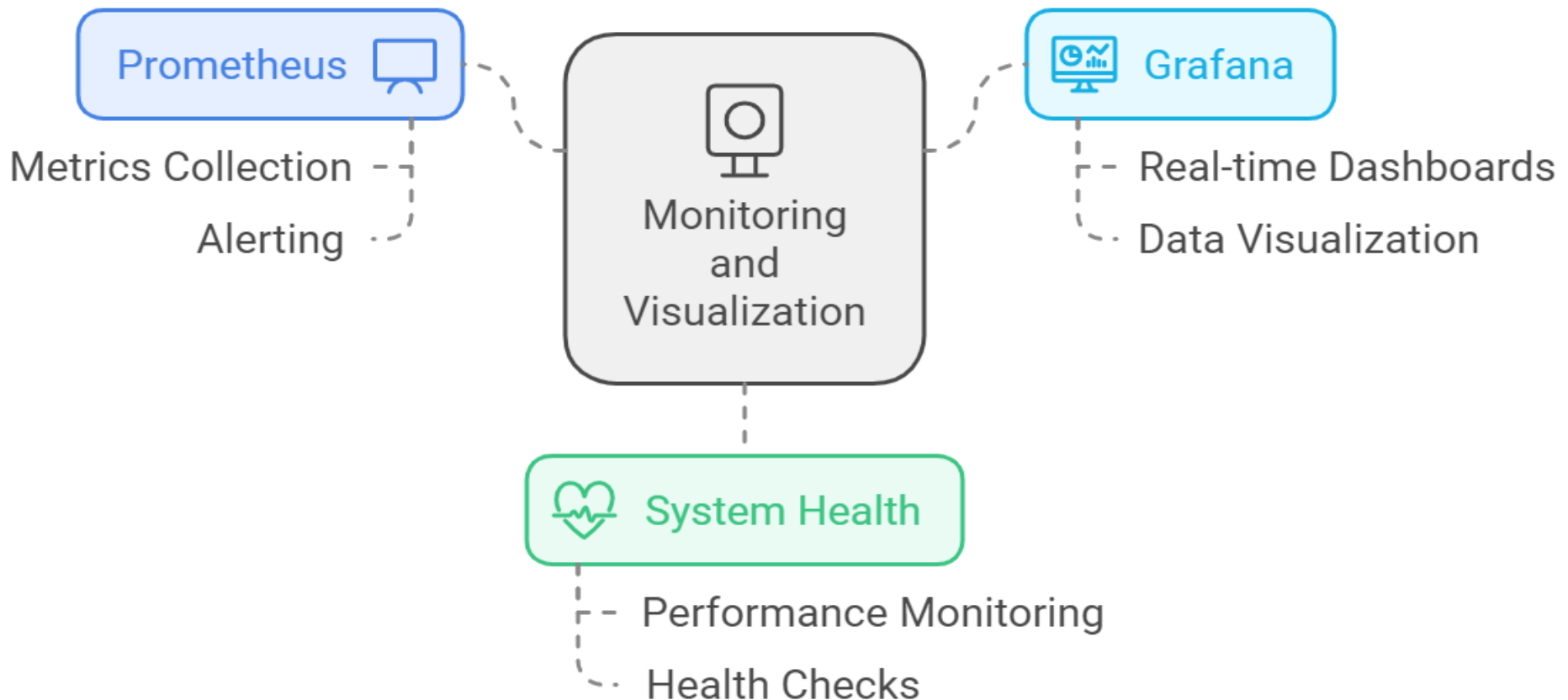
Application Deployment to Kubernetes

The application is deployed to both staging and production environments in Kubernetes using the both imperative and declarative methodologies.



Monitoring using Prometheus and Grafana

Prometheus monitors the application, and Grafana provides real-time dashboards to visualize performance metrics. This ensures the system's health and performance.



Conclusion

CI/CD pipelines with modern tools like GitHub Actions, Docker, Terraform, AWS, Ansible, Kubernetes, Prometheus, and Grafana ensure rapid, reliable, and consistent software delivery.

